



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2012
ASSESSMENT TASK 2

Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes
 - Working time – 55 minutes
 - Write in the booklet provided
 - Write using blue or black pen
 - Board approved calculators may be used
 - All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
 - Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Weiss
- ☐ Mr Ireland
- ☐ Mr Fletcher

Student Number:

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	Total	Total
Mark	$\frac{4}{4}$	$\frac{2}{2}$	$\frac{9}{9}$	$\frac{13}{13}$	$\frac{5}{5}$	$\frac{7}{7}$	$\frac{40}{40}$	$\frac{100}{100}$

Question 1 (4 marks)**Marks**

Question 1 is the **objective response** section of the paper. Each question is worth **1 mark**.

You must select the correct response to each of the following four questions and shade the appropriate box on the supplied answer sheet.

- (i) An ellipse has eccentricity $e = \frac{\sqrt{5}}{3}$. Its equation could be:
- (A) $\frac{x^2}{75} + \frac{y^2}{50} = 1$ (B) $\frac{x^2}{45} + \frac{y^2}{20} = 1$ (C) $\frac{x^2}{48} + \frac{y^2}{32} = 1$ (D) $\frac{x^2}{45} + \frac{y^2}{25} = 1$
- (ii) A tangent is drawn to the rectangular hyperbola $xy = 4$ at the point $(2, 2)$ on it. The tangent cuts the axes at points A and B . The length of AB is:
- (A) $2\sqrt{2}$ (B) $\sqrt{2}$ (C) $4\sqrt{2}$ (D) $\frac{\sqrt{2}}{2}$
- (iii) The asymptotes of the hyperbola $4y^2 - x^2 = 4$ are:
- (A) $y = \pm \frac{x}{4}$ (B) $y = \pm 2x$ (C) $y = \pm 4x$ (D) $y = \pm \frac{x}{2}$
- (iv) The conic $\frac{x^2}{25} + \frac{y^2}{16 - \lambda} = 1$ will be an ellipse with foci on the x -axis provided that λ lies in which of the following ranges of values:
- (A) $\lambda < 16$ (B) $\lambda > -9$ (C) $0 < \lambda < 16$ (D) $-9 < \lambda < 16$

Question 2 (2 marks)

Now open your answer booklet and write your solutions.

Find real numbers a , b and c such that $\frac{1}{x(4+x^2)} \equiv \frac{a}{x} + \frac{bx+c}{4+x^2}$ 2

Question 3 (9 marks)

Consider the rectangular hyperbola $xy = 16$. A chord, ST , is formed by joining the points

$S(4s, \frac{4}{s})$ and $T(4t, \frac{4}{t})$.

- (i) Show that the equation of ST is $x + sty = 4(s+t)$. 2
- (ii) It is given that ST passes through the point $(8, 8)$.
Show that $2st = s + t - 2$. 1

- (iii) Show that the tangents at S and T meet at $R(\frac{8st}{s+t}, \frac{8}{s+t})$. 4
- (iv) Find the equation of the locus of R as S and T vary on the hyperbola. 2

Question 4 (13 marks) Start a new page.

The hyperbola H has equation $\frac{x^2}{4} - \frac{y^2}{7} = 1$.

- (i) Write down the eccentricity, e , the coordinates of the foci, S and S' , the equations of its directrices, and the equations of its asymptotes. 4
- (ii) Sketch the hyperbola, showing all the details of part (i). 2
- (iii) Let P be the arbitrary point $(2 \sec \theta, \sqrt{7} \tan \theta)$.
Show that P lies on H , and prove that the tangent to H at P has equation
$$\frac{x \sec \theta}{2} - \frac{y \tan \theta}{\sqrt{7}} = 1.$$
 4
- (iv) This tangent cuts the asymptotes at L and M .
Prove that $LP = PM$. 3

Question 5 (5 marks) Start a new page.

An ellipse E has equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

- (i) Show that the tangent at any point (x_1, y_1) on E is given by
$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1.$$
 2
- (ii) Suppose that point P is an extremity of a latus rectum of E .
A tangent is drawn at P , meeting the x -axis at A and the y -axis at B .
Show that $\frac{PA}{PB} = \frac{1-e^2}{e^2}$. 3
- [You may find it helpful to draw a diagram showing A, B, P , and other relevant points.]

Please turn the page for the last question →

Question 6 (7 marks) *Start a new page.*

- (i) Show that the condition for the line $y = mx + c$ to be tangent to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{is that } c^2 = a^2 m^2 + b^2. \quad 3$$

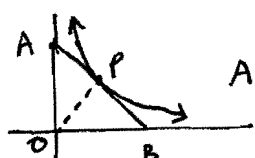
- (ii) Hence, or otherwise, prove that the pair of tangents from the point (3,4)

$$\text{to the ellipse } \frac{x^2}{16} + \frac{y^2}{9} = 1 \text{ are at right angles to each other.} \quad 4$$

END OF EXAM

Q1

(i) B [Reason: $\frac{b^2}{a^2} = 1 - e^2 = \frac{4}{9}$]

(ii) C [Reason:  $AB = 2 \times OP$]

(iii) D [Reason: $\frac{y^2}{1} - \frac{x^2}{4} = 1$, $y = \pm \frac{b}{a}x = \pm \frac{1}{2}x$]

(iv) D [Reason: $0 < 16 - \lambda < 25 \therefore -9 < \lambda < 16$]

/ 4

Q2

$$\frac{a}{x} + \frac{bx+c}{4+x^2} \equiv \frac{4a+ax^2+bx^2+cx}{x(4+x^2)}$$

$$\therefore 4a=1, a=-b, c=0$$

$$\text{Thus, } a = \frac{1}{4}, b = -\frac{1}{4}, c=0.$$

/ 2

Q3

(i) $m_{st} = \frac{\frac{4}{t} - \frac{4}{s}}{4t - 4s} = \frac{\frac{4s-4t}{st}}{4t-4s} = \frac{-1}{st}$

$$\therefore \text{chord equation is } y - \frac{4}{s} = -\frac{1}{st}(x-4s)$$

$$\therefore sty - 4t = -x + 4s$$

$$\therefore x + sty = 4(s+t), \text{ as required.}$$

(ii) Through (8,8) $\therefore 8 + 8st = 4(s+t)$

$$\therefore 2 + 2st = s+t$$

$$\therefore 2st = s+t - 2$$

Q3 - continued

(iii) From part (i),

let $t \rightarrow s$.Thus tangent at S will be $x + s^2 y = 4(2s)$ ✓✓

$$\therefore x + s^2 y = 8s \quad \text{---- (1)}$$

Similarly, tangent at T is:

$$x + t^2 y = 8t \quad \text{---- (2)}$$

[Alternatively, if $xy = 16$, $\therefore y \cdot 1 + x \cdot \frac{dy}{dx} = 0$

$$\therefore \frac{dy}{dx} = -\frac{y}{x} = \frac{-\frac{4}{s}}{4s} \quad \text{at S}$$

$$= -\frac{1}{s^2} \quad (\checkmark)$$

$$\therefore \text{tangent at S is } y - \frac{4}{s} = -\frac{1}{s^2} (x - 4s)$$

$$\therefore x + s^2 y = 8s \quad (\checkmark)$$

$$\text{Likewise at T, } x + t^2 y = 8t. \quad]$$

Solving simultaneously, $y(s^2 - t^2) = 8(s - t)$

$$\therefore y = \frac{8}{s+t} \quad (\text{as } s \neq t) \quad \checkmark$$

$$\text{Now } t^2 x + t^2 s^2 y = 8st^2 \quad (\text{multiplying (1) by } t^2)$$

$$\& \quad s^2 x + s^2 t^2 y = 8ts^2 \quad (\text{mult. (2) by } s^2)$$

$$\text{Subtracting, } x(t^2 - s^2) = 8st(t - s)$$

$$\therefore x = \frac{8st}{s+t} \quad (\text{as } s \neq t) \quad \checkmark$$

$$\therefore R = \left(\frac{8st}{s+t}, \frac{8}{s+t} \right) \text{ as required.}$$

Q3 - continued

(iv) Since $R = \left(\frac{8st}{s+t}, \frac{8}{s+t} \right)$,

we see $x = sty$.

Using part (ii), $st = \frac{s+t}{2} - 1$
 $= \frac{4}{y} - 1$

$\therefore x = \left(\frac{4}{y} - 1 \right) y = 4 - y$

Thus $x + y - 4 = 0$ is the equation of the locus of R . ✓

Note however that since $s \neq 0$ and $t \neq 0$,
 $\therefore x \neq 0$.

Likewise, since $y = \frac{8}{s+t}$, $\therefore y \neq 0$. ✓

So locus is $x + y - 4 = 0$ ($x \neq 0, y \neq 0$) ✓

[i.e. excluding the points $(0, 4), (4, 0)$].

[Note also that s cannot take the value 1 or -1, otherwise $s+t=0$, i.e. the tangents do not intersect.]

Q4

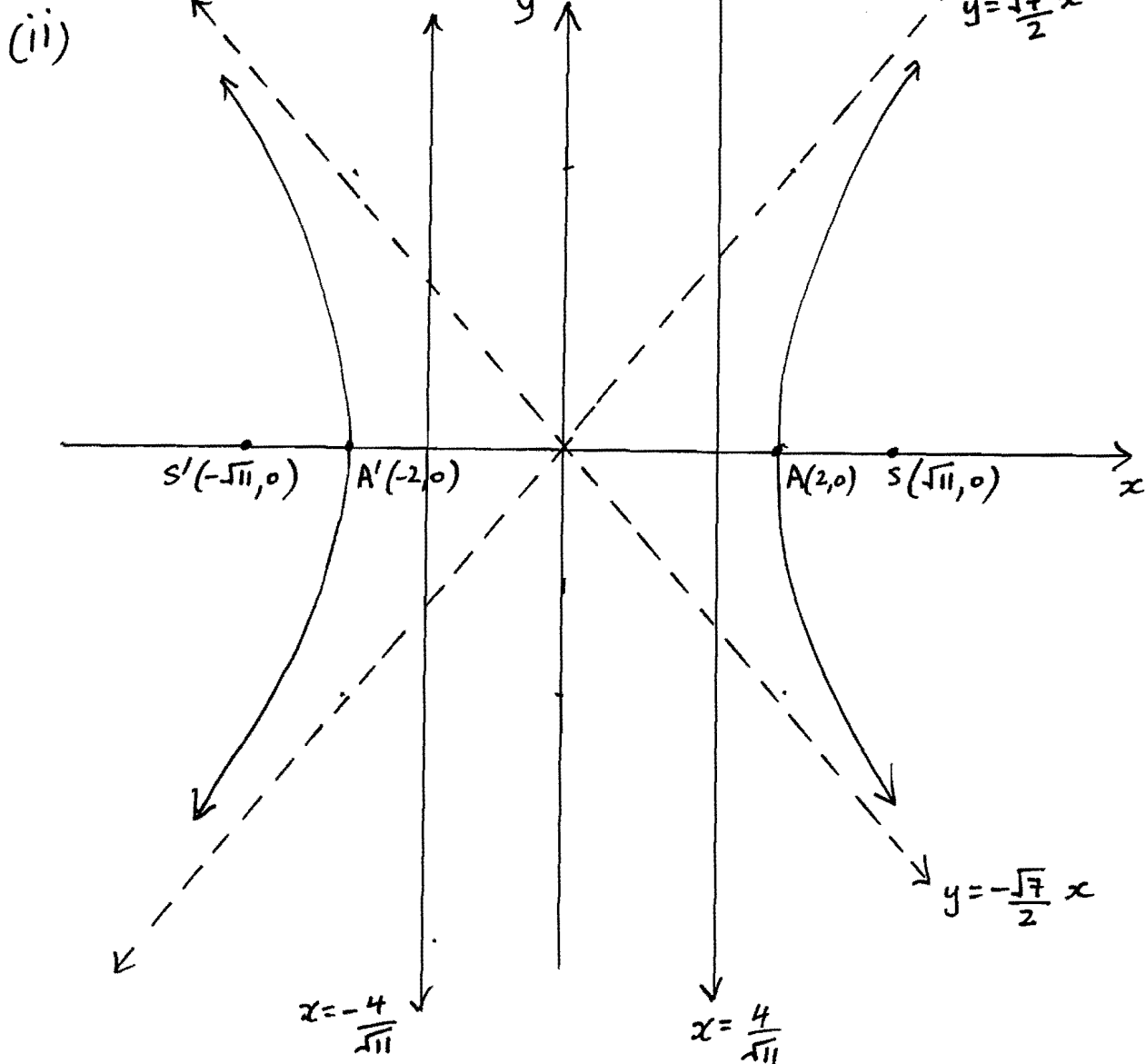
$$(i) \frac{x^2}{4} - \frac{y^2}{7} = 1 \quad \therefore a^2 = 4, b^2 = 7$$

$$\text{Since } b^2 = a^2(e^2 - 1) \quad \therefore e^2 = \frac{11}{4} \quad \therefore e = \frac{\sqrt{11}}{2}$$

$$S = (ae, 0) = (\sqrt{11}, 0), \quad S' = (-\sqrt{11}, 0)$$

$$\text{Directrices are } x = \pm \frac{a}{e} = \pm \frac{2}{\frac{\sqrt{11}}{2}} \quad \therefore x = \pm \frac{4}{\sqrt{11}}$$

$$\text{Asymptotes are } y = \pm \frac{b}{a}x \quad \therefore y = \pm \frac{\sqrt{7}}{2}x$$



Q4 - continued

$$\begin{aligned}
 \text{(iii) At } P, \quad \frac{x^2}{4} - \frac{y^2}{7} &= \frac{4\sec^2\theta}{4} - \frac{7\tan^2\theta}{7} \\
 &= \sec^2\theta - \tan^2\theta \\
 &= 1 \quad \therefore P \text{ is on } H
 \end{aligned}$$

Differentiating, $\frac{2x}{4} - \frac{2y}{7} \cdot \frac{dy}{dx} = 0$

$$\begin{aligned}
 \therefore \frac{dy}{dx} &= \frac{7x}{4y} \\
 &= \frac{\sqrt{7}}{2} \frac{\sec\theta}{\tan\theta} \text{ at } P.
 \end{aligned}$$

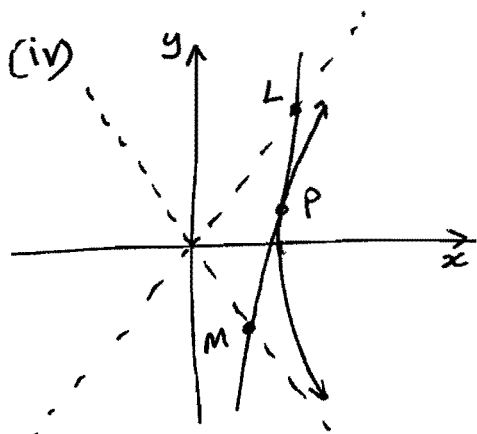
$\therefore$ tangent at P is :

$$y - \sqrt{7} \tan\theta = \frac{\sqrt{7} \sec\theta}{2 \tan\theta} (x - 2 \sec\theta)$$

$$\therefore 2 \tan\theta y - 2 \sqrt{7} \tan^2\theta = \sqrt{7} \sec\theta x - 2 \sqrt{7} \sec^2\theta$$

$$\begin{aligned}
 \therefore \sqrt{7} \sec\theta x - 2 \tan\theta y &= 2 \sqrt{7} (\sec^2\theta - \tan^2\theta) \\
 &= 2 \sqrt{7}
 \end{aligned}$$

$$\therefore \frac{\sec\theta}{2} x - \frac{\tan\theta}{\sqrt{7}} y = 1, \text{ as required.}$$



At L: $\frac{x \sec\theta}{2} - \frac{y \tan\theta}{\sqrt{7}} = 1$

and $y = \frac{\sqrt{7}}{2} x$

$$\therefore \frac{x \sec\theta}{2} - \frac{\tan\theta}{\sqrt{7}} \cdot \frac{\sqrt{7}}{2} x = 1$$

$$\begin{aligned}
 \therefore x &= \frac{2}{\sec\theta - \tan\theta} \\
 &= \frac{2(\sec\theta + \tan\theta)}{\sec^2\theta - \tan^2\theta}
 \end{aligned}$$

$$= 2(\sec\theta + \tan\theta)$$

$$\therefore y = \frac{\sqrt{7}}{2} x = \sqrt{7}(\sec\theta + \tan\theta)$$

✓ for L

Q4 - continued

(iv) - continued

At M, $y = -\frac{\sqrt{7}}{2}x$

$$\therefore \frac{x \sec \theta}{2} - \frac{\tan \theta}{\sqrt{7}} \cdot -\frac{\sqrt{7}}{2}x = 1$$

$$\therefore x = \frac{2}{\sec \theta + \tan \theta} = 2(\sec \theta - \tan \theta)$$

$$\therefore y = -\sqrt{7}(\sec \theta - \tan \theta)$$

✓ for M

To prove $LP = PM$, just show P is the midpoint of LM:

midpoint of LM has x-coordinate:

$$x = \frac{2(\sec \theta + \tan \theta) + 2(\sec \theta - \tan \theta)}{2}$$

$$= 2\sec \theta$$

It has y coordinate:

$$y = \frac{\sqrt{7}(\sec \theta + \tan \theta) - \sqrt{7}(\sec \theta - \tan \theta)}{2}$$

$$= \sqrt{7} \tan \theta$$

$$\text{i.e. midpoint of LM} = (2\sec \theta, \sqrt{7} \tan \theta) = P$$

$$\therefore LP = PM$$

✓

[Alternatively, candidates may use the distance formula to calculate LP and PM].

Q5 (i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\therefore \frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\therefore \text{at } P(x_1, y_1), \quad \frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1}$$

$\therefore$ tangent is:

$$y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1)$$

$$a^2 y_1 y - a^2 y_1^2 = -b^2 x_1 x + b^2 x_1^2$$

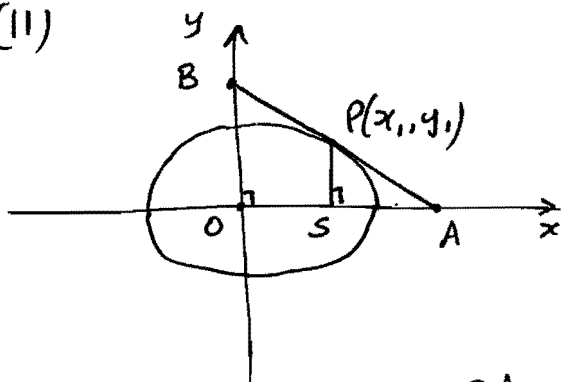
$$\therefore b^2 x_1 x + a^2 y_1 y = b^2 x_1^2 + a^2 y_1^2$$

$$\therefore \frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$$

$$= 1 \quad (\text{as } P \text{ is on } E)$$

... as required.

(ii)



At A, $y = 0$

$$\therefore x = \frac{a^2}{x_1}$$

$$= \frac{a^2}{ae}$$

as P on
Latus Rectum

$$\therefore x = \frac{a}{e}$$

$$\text{Then } \frac{PA}{PB} = \frac{SA}{OS}$$

(ratios of intercepts on // lines)

$$= \frac{\frac{a}{e} - ae}{ae} = \frac{\frac{1}{e^2} - 1}{\frac{1}{e^2}} = \frac{1 - e^2}{e^2}$$

... as required.

[Q5] - ALTERNATIVELY... A LONGER SOLUTION IS :-

$$\text{At } P, x = ae \quad \therefore \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} = 1 - e^2$$

$$\begin{aligned} \therefore y^2 &= b^2(1 - e^2) \\ &= a^2(1 - e^2) \cdot (1 - e^2) \\ \therefore y &= \pm a(1 - e^2) \end{aligned}$$

$$\therefore P = [ae, a(1 - e^2)]$$

$$\text{At } A, y = 0 \quad \therefore x = \frac{a^2}{x_1} = \frac{a^2}{ae} = \frac{a}{e}$$

$$\therefore A = \left(\frac{a}{e}, 0\right)$$

$$\begin{aligned} \text{At } B, x = 0 \quad \therefore y &= \frac{b^2}{y_1} = \frac{b^2}{a(1 - e^2)} = \frac{a^2(1 - e^2)}{a(1 - e^2)} \\ &= a \end{aligned}$$

$$\therefore B = (0, a)$$

$$\begin{aligned} \text{Then } PA &= \sqrt{\left(ae - \frac{a}{e}\right)^2 + a^2(1 - e^2)^2} \\ &= a(1 - e^2) \sqrt{\frac{1}{e^2} + 1} \\ &= \frac{a(1 - e^2)}{e} \sqrt{1 + e^2} \end{aligned}$$

$$\begin{aligned} \text{And } PB &= \sqrt{a^2e^2 + (a - a(1 - e^2))^2} \\ &= \sqrt{a^2e^2 + a^2e^4} \\ &= ae \sqrt{1 + e^2} \end{aligned}$$

$$\therefore \frac{PA}{PB} = \frac{\frac{a(1 - e^2) \sqrt{1 + e^2}}{e}}{ae \sqrt{1 + e^2}} = \frac{1 - e^2}{e^2}$$

... as required.]

Q6 (i) Where line meets ellipse, we have

$$\frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$$\therefore b^2 x^2 + a^2 (mx+c)^2 = a^2 b^2$$

$$b^2 x^2 + a^2 m^2 x^2 + 2mc a^2 x + a^2 c^2 = a^2 b^2$$

$$\therefore (b^2 + a^2 m^2) x^2 + 2a^2 m c x + (a^2 c^2 - a^2 b^2) = 0$$

For tangent, $\Delta = 0$

$$\therefore 4a^4 m^2 c^2 - 4(b^2 + a^2 m^2)(a^2 c^2 - a^2 b^2) = 0$$

$$\therefore a^2 m^2 c^2 - (b^2 + a^2 m^2)(c^2 - b^2) = 0 \quad \text{as } a \neq 0$$

$$\therefore a^2 m^2 c^2 - (b^2 c^2 - b^4 + a^2 m^2 c^2 - a^2 m^2 b^2) = 0$$

$$\therefore -c^2 + b^2 + a^2 m^2 = 0 \quad \text{as } b^2 \neq 0$$

$$\therefore c^2 = a^2 m^2 + b^2 \quad \dots \text{as required.}$$

(ii) From (i), $y = mx + c$ is a tangent to

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \quad \text{if } c^2 = 16m^2 + 9$$

$$\text{i.e. } c = \pm \sqrt{16m^2 + 9}$$

$$\therefore \text{tangents are } y = mx \pm \sqrt{16m^2 + 9}$$

If tangents pass through (3, 4) we have:

$$4 = 3m \pm \sqrt{16m^2 + 9}$$

$$\therefore 4 - 3m = \pm \sqrt{16m^2 + 9}$$

$$\therefore 16 - 24m + 9m^2 = 16m^2 + 9$$

$$\therefore 7m^2 + 24m - 7 = 0$$

Q6 - continued

The equation $7m^2 - 24m - 7 = 0$
has 2 solutions for m , corresponding to
the 2 tangents.

The product of the solutions $= \frac{-7}{7} = -1$ ✓

$$\text{i.e. } m_1 \times m_2 = -1$$

i.e. the tangents are perpendicular.

[Alternatively, candidates can solve the equation,
getting $m = \frac{24 \pm \sqrt{576 + 196}}{14}$

$$= \frac{24 \pm \sqrt{772}}{14}$$

$$\therefore m_1 \cdot m_2 = \frac{24 + \sqrt{772}}{14} \cdot \frac{24 - \sqrt{772}}{14}$$

$$= \frac{24^2 - 772}{196}$$

$$= -1$$

$\therefore$ tangents are perpendicular.]